PROBLEM CL1

Airfoil Wake: Streamline Displacement and Drag

Original statement · CL1.pdf

Control volume | Linear momentum | Wake | Drag

OFFICIAL PROBLEM STATEMENT

The experimental tests of an airfoil in a wind tunnel indicate that the velocity profile upstream remains uniform, whereas behind the airfoil there exists a wake region with significant velocity deficit whose thickness increases downstream, resulting in the flow field sketched in the figure of CL1.pdf. For the airfoil tested, the velocity measurements at a given downstream section reveal that the velocity in the wake varies linearly from the value $u = U/2$ at the centerline to reach the value $u = U$ at the edge of the wake, located at $y = y_0$. To analyze the problem, consider a control volume limited laterally by the streamlines that extend from $x \to -\infty$, $y = \pm y_{-\infty}$ ($y_{-\infty} < y_0$) to $x = x_0$, $y = \pm y_0$ and internally by the surface of the airfoil. Assuming that the air behaves as an incompressible fluid with density ρ and that viscous stresses are negligible both in the wake and along the bounding streamlines, obtain the value of $y_{-\infty}$ as a function of y_0, as well as the value of the drag on the airfoil as a function of ρ, y_0 and U. In the analysis, assume that on the outer boundary of the control volume the pressure differences from the ambient value p_∞ are negligible, and so is the effect of gravity on the air flow.

Source: CL1.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eqs. 3.10-3.11, integral momentum balance Eqs. 3.35-3.36, force of the fluid on a body Eq. 3.32.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Degrees of Freedom and Data

- Steady flow ($\partial/\partial t = 0$), planar (everything is computed per unit span in the direction z normal to the figure), incompressible (constant ρ), no gravity.
- Viscous stresses are negligible in the wake and on the bounding streamlines (given); pressure differences $p - p_\infty$ are negligible on the whole outer boundary of the control volume (given). Pressure and viscous stresses on the airfoil surface are **not** neglected: they are the drag.

- Wake profile at the section $x = x_0$, symmetric about $y = 0$: $v_x = \frac{U}{2}\left(1 + \frac{|y|}{y_0}\right)$ for $|y| \leq y_0$ and $v_x = U$ for $|y| > y_0$. The only unknowns are the scalar $y_{-\infty}$ and the drag D.

Symbol	Meaning	SI unit
U	uniform upstream velocity	m/s
y_0	half-width of the wake at $x = x_0$	m
$y_{-\infty}$	half-width of the stream tube far upstream	m
ρ	air density	kg/m^3
D	drag per unit span	N/m

Dimensional check of the unknown: $[\rho U^2 y_0] = (\mathrm{kg/m^3})(\mathrm{m^2/s^2})\,\mathrm{m} = \mathrm{kg/s^2} = \mathrm{N/m}$, a force per unit length, as required for a planar problem.

PHASE 2

Coordinates, Frames and Changes of Variable

- Inertial frame fixed to the airfoil and the wind tunnel, Cartesian basis $(\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z)$ with $\mathbf{e}_x$ along the incoming stream and $\mathbf{e}_z = \mathbf{e}_x \wedge \mathbf{e}_y$. The control volume is fixed in this frame, so $\mathbf{v}_c = 0$ in the universal balances (Eqs. 3.10, 3.35 reduce to Eqs. 3.11, 3.36).
- Boundary of the control volume: Σ_u (section far upstream, $|y| < y_{-\infty}$, $\mathbf{n} = -\mathbf{e}_x$), Σ_d (section $x = x_0$, $|y| < y_0$, $\mathbf{n} = +\mathbf{e}_x$), Σ_l (the two bounding streamlines), and Σ_w (airfoil surface, with $\mathbf{n}$ pointing out of the fluid, i.e. into the airfoil).
- The force of the fluid on the airfoil is $\mathbf{F} = \int_{\Sigma_w} (-p\,\mathbf{n}_w + \bar{\bar{\tau}}' \cdot \mathbf{n}_w)\, d\sigma$ with $\mathbf{n}_w = -\mathbf{n}$ pointing out of the airfoil (Eq. 3.32). The force of the airfoil on the fluid is therefore $-\mathbf{F}$.
- Change of variable used in the integral of the wake: $s = 1 + y/y_0$, $ds = dy/y_0$, so that $y \in [0, y_0]$ maps to $s \in [1, 2]$.

PHASE 3

Step-by-Step Derivation

3.1 Mass balance: the width $y_{-\infty}$ of the incoming stream tube

Why this tool: the control volume is fixed and the flow is steady, so the integral continuity equation (Notes Eq. 3.11) reduces to $\int_{\Sigma_c} \rho\, \mathbf{v} \cdot \mathbf{n}\, d\sigma = 0$. It is the only relation that links the unknown $y_{-\infty}$ to the measured profile.

On Σ_l (streamlines) and on Σ_w (impermeable wall) $\mathbf{v} \cdot \mathbf{n} = 0$, so only Σ_u and Σ_d contribute. On Σ_u, $\mathbf{v} \cdot \mathbf{n} = -U$ over a width $2y_{-\infty}$:

$$\int_{\Sigma_u} \rho\,\mathbf{v}\cdot\mathbf{n}\,d\sigma = -\rho U\,(2y_{-\infty}).$$

On Σ_d, $\mathbf{v}\cdot\mathbf{n} = v_x$ and, by symmetry, the integral is twice the integral over $0 \le y \le y_0$. The antiderivative of $1 + y/y_0$ is $y + y^2/(2y_0)$, and Barrow's rule between $y = 0$ and $y = y_0$ gives:

$$\int_{\Sigma_d} \rho\,\mathbf{v}\cdot\mathbf{n}\,d\sigma = 2\rho\int_0^{y_0} \frac{U}{2}\left(1 + \frac{y}{y_0}\right) dy = \rho U\left[y + \frac{y^2}{2y_0}\right]_0^{y_0} = \rho U\left(y_0 + \frac{y_0}{2}\right) = \frac{3}{2}\rho U y_0.$$

Adding both contributions and setting the sum to zero:

$$-2\rho U y_{-\infty} + \frac{3}{2}\rho U y_0 = 0 \implies \boxed{y_{-\infty} = \frac{3}{4}\,y_0.}$$

3.2 Linear momentum balance: the drag

Why this tool: the force of the fluid on a body is not accessible from the (unknown) pressure and shear on its surface, but it appears in the integral momentum balance (Notes Eq. 3.36, steady, fixed control volume, no gravity) as the only unknown surface term:

$$\int_{\Sigma_c} \rho\,\mathbf{v}\,(\mathbf{v}\cdot\mathbf{n})\,d\sigma = \int_{\Sigma_c} \left(-p\,\mathbf{n} + \bar{\bar{\tau}}'\cdot\mathbf{n}\right) d\sigma.$$

Because Σ_c is a closed surface, $\int_{\Sigma_c} p_\infty \mathbf{n}\,d\sigma = 0$, and p may be replaced by $p - p_\infty$ on the right-hand side. On Σ_u, Σ_d and Σ_l the difference $p - p_\infty$ and the viscous stresses are negligible (hypothesis), so the only surviving term is the one on Σ_w, which equals the force of the airfoil on the fluid, $-\mathbf{F}$. The x component gives the drag $D = F_x$:

$$D = -\int_{\Sigma_u+\Sigma_d+\Sigma_l} \rho\, v_x\, (\mathbf{v}\cdot\mathbf{n})\, d\sigma.$$

On Σ_l, $\mathbf{v}\cdot\mathbf{n} = 0$. On Σ_u, $v_x = U$ and $\mathbf{v}\cdot\mathbf{n} = -U$, hence with 3.1:

$$\int_{\Sigma_u} \rho\, v_x (\mathbf{v}\cdot\mathbf{n})\, d\sigma = -\rho U^2\,(2y_{-\infty}) = -\frac{3}{2}\rho U^2 y_0.$$

On Σ_d, with the substitution $s = 1 + y/y_0$, $dy = y_0\, ds$, and Barrow's rule between $s = 1$ and $s = 2$:

$$\int_{\Sigma_d} \rho\, v_x^2\, d\sigma = 2\rho\,\frac{U^2}{4}\int_0^{y_0}\left(1+\frac{y}{y_0}\right)^2 dy = \frac{\rho U^2}{2}\,y_0\int_1^2 s^2\, ds = \frac{\rho U^2 y_0}{2}\left[\frac{s^3}{3}\right]_1^2 = \frac{\rho U^2 y_0}{2}\cdot\frac{8-1}{3} = \frac{7}{6}\rho U^2 y_0.$$

Therefore:

$$D = -\left(-\frac{3}{2}+\frac{7}{6}\right)\rho U^2 y_0 = -\left(-\frac{9}{6}+\frac{7}{6}\right)\rho U^2 y_0 \;\Longrightarrow\; \boxed{D = \frac{1}{3}\,\rho\, U^2\, y_0} \quad \text{per unit span, in the direction } +\mathbf{e}_x.$$

For a wake symmetric about $y = 0$ (a symmetric airfoil at zero angle of attack, which the statement implies by giving a symmetric profile) the y component of the same balance gives $F_y = 0$: there is no lift.

PHASE 4

Interpretation, Limits and Dimensional Check

- Physical picture: the wake carries less mass flux ($\frac{3}{2}\rho U y_0$) than the undisturbed stream would through the same width ($2\rho U y_0$); the missing flux is the volume displaced sideways by the wake, which is why the bounding streamlines are at $y_{-\infty} = \frac{3}{4}y_0 < y_0$ far upstream. The momentum deficit left in the wake is the drag.
- Limit $y_0 \to 0$: no wake, $y_{-\infty} \to 0$ and $D \to 0$, consistent with d'Alembert's paradox for a frictionless attached flow.

- Dimensions (SI): $[y_{-\infty}] = [y_0] = \mathrm{m}$; $[D] = [\rho U^2 y_0] = \mathrm{N/m}$.
- Numerical cross-check (midpoint quadrature with 2×10^5 cells of the defining integrals, $U = 1$ m/s, $\rho = 1.2$ kg/m^3, $y_0 = 0.35$ m): $y_{-\infty}/y_0 = 0.75000$ and $D = 0.140000$ N/m, equal to $\rho U^2 y_0/3 = 0.14$ N/m.
- Agreement with the official solution: CL1.pdf obtains $y_{-\infty} = \frac{3}{4} y_0$ and $\mathbf{F}_{\text{airfoil}} = \frac{1}{3}\rho U^2 y_0\, \mathbf{e}_x$, the same results.

Extension (beyond the official solution)

With the momentum thickness $\theta = \int_{-y_0}^{y_0} \frac{v_x}{U}\left(1 - \frac{v_x}{U}\right) dy = 2y_0 \int_0^1 \frac{1-\sigma^2}{4}\, d\sigma = \frac{y_0}{3}$ (substituting $\sigma = y/y_0$), the result reads $D = \rho U^2 \theta$: the classical momentum-deficit formula for the drag from a wake survey.